Forecast of El Niño

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Contents

Introduction

El Niño

- El Niño is the warm phase of ENSO in the tropical Pacific: a major source of interannual climate variability.
- Social and environmental impacts: rainfall, droughts, floods, economy.

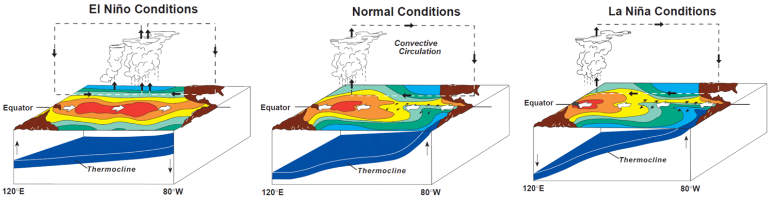


Figure: El Niño Southern Oscillation (ENSO) cycle.

Project Question:
How predictable is El Niño? How far ahead can we forecast Niño-3.4 anomalies with simple statistical models?

Data

Dataset

- **Variables:** sea surface temperature (SST)
  mean surface level pressure (MSLP)
- **Temporal resolution:** monthly mean from 1981 to 2021
- **Spatial resolution:** 1 degree X 1 degree

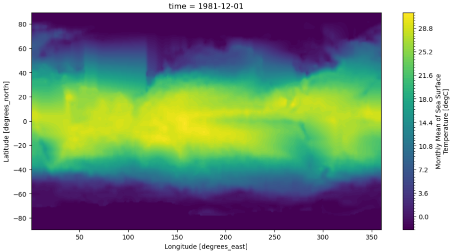


Figure: SST at time 0.

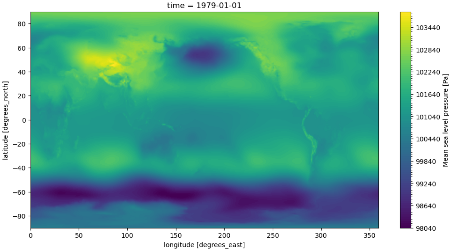
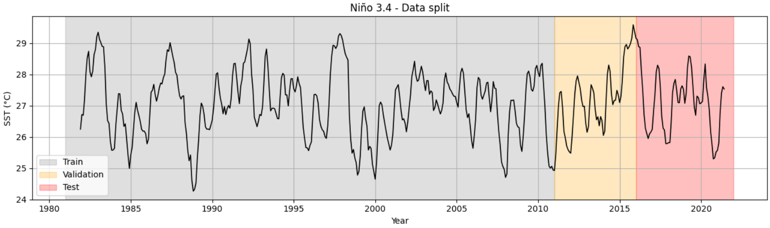


Figure: MSLP at time 0.

Data split

- Train → from 1981 to 2010
- Validation → from 2011 to 2015
- Test → from 2016 to 2021



The training slice was also used to compute the **base period climatology** (30 years)

EOF/PCA

EOF/PCA

Empirical Orthogonal Function (EOF) analysis is used to decompose climate data into dominant spatial patterns and their associated temporal variability.

For a climate variable X, the anomaly field can be written as:

$$X(x,y,t) = \sum_{k=1}^{K} \mathrm{PC}_k(t)\,\mathrm{EOF}_k(x,y)$$

- EOF_k are orthogonal spatial patterns ordered by explained variance
- PC_k are the associated time series describing their temporal evolution.

EOF/PCA

The analysis is restricted to the Tropical Pacific region for both anomaly fields.

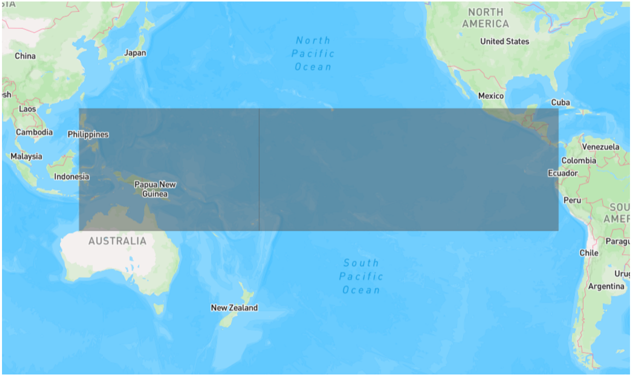
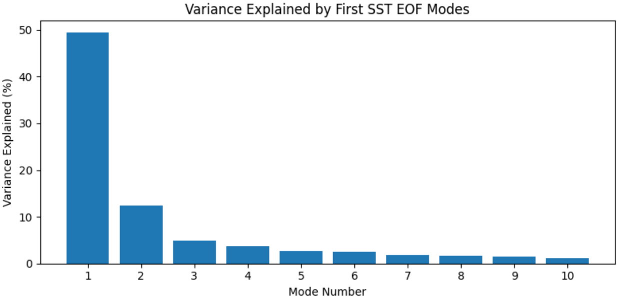


Figure: Tropical Pacific Region

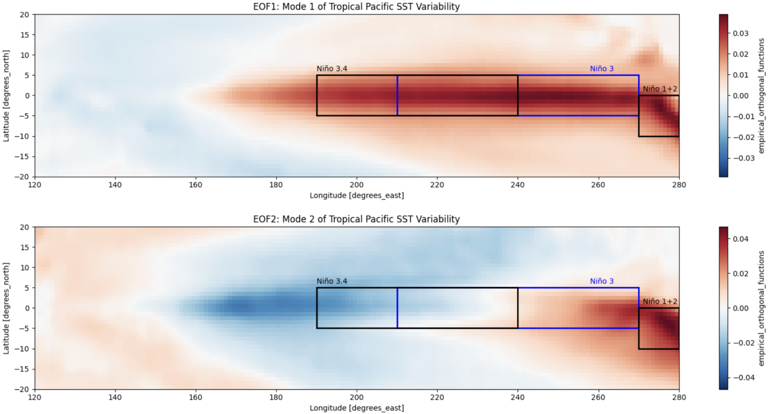
EOF/PCA - SST Anomaly

Fraction of total SST-anomaly variance captured by each EOF:



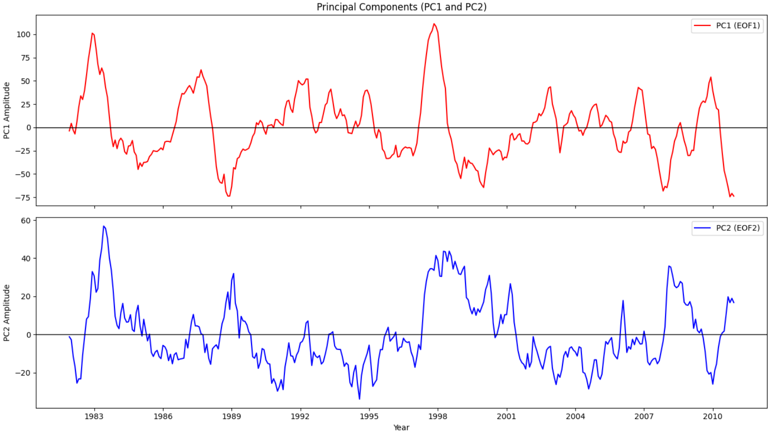
- The first 2 SST EOF modes explain 61.8% of the variance.
- The first 10 SST EOF modes explain 81.8% of the variance.

EOFs - SST Anomaly



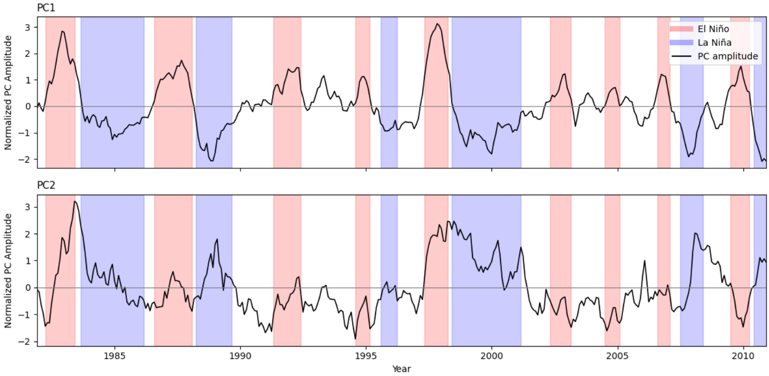
- Which locations tend to vary together on the grid

PCs - SST Anomaly



- How strongly the climate field at each time follows that EOF pattern.

PCs - SST Anomaly



- How strongly the climate field at each time follows that EOF pattern.

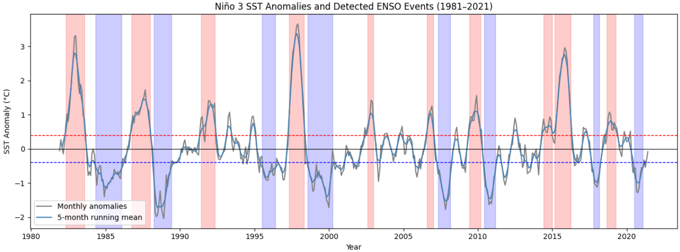
Characterizing El Niño

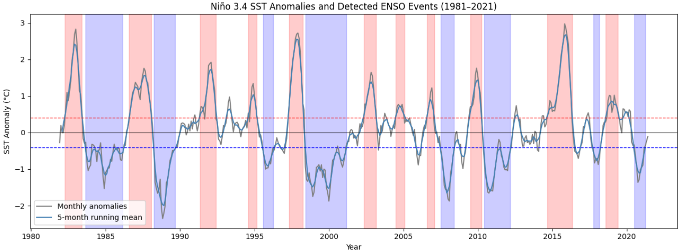
Characterizing El Niño events from data

Definition: An El Niño event occurs when the 5-month running mean of sea surface temperature (SST) anomalies in the Niño 3.4 region (5° N-5° S, 120°-170° W) remains above $+0.4$ °C for at least six consecutive months. [1]

- Niño 3 index;
- Niño 3.4 index;
- Southern oscillation Index (SOI).

Niño 3 and Niño 3.4

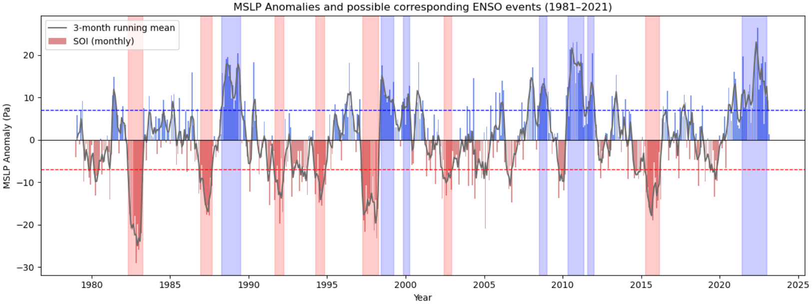




Southern Oscillation Index (SOI)

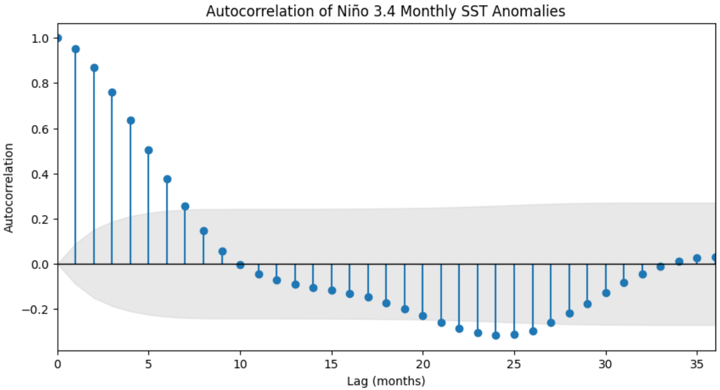
The atmospheric part of ENSO.
The pressure difference between the western and eastern tropical Pacific.

- SOI < -7 $\rightarrow$ low pressure in the east $\rightarrow$ often indicates El Niño.
- SOI $> +7$ $\rightarrow$ low pressure in the west $\rightarrow$ often indicates La Niña.



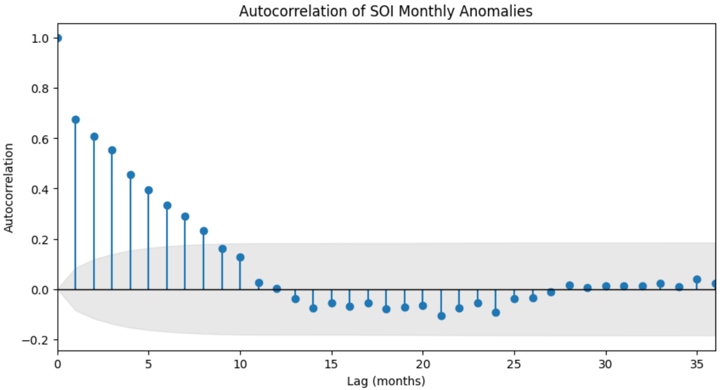
Correlation Analysis

Autocorrelation - Niño 3.4



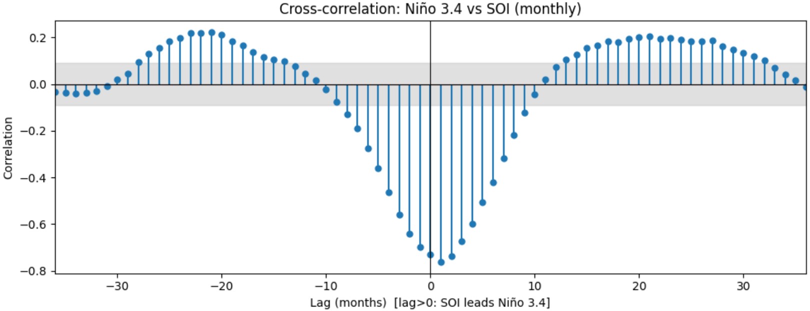
- Strong positive autocorrelation during the first 3-6 months

Autocorrelation - SOI



- Relevant positive autocorrelation during the first 6 months

Cross-Correlation



- Strong anti-correlation between SOI and Niño 3.4 at short lags
- Peak correlation at lag 1 indicates that SOI slightly leads Niño 3.4

Forecasting

Models: Time series forecasting

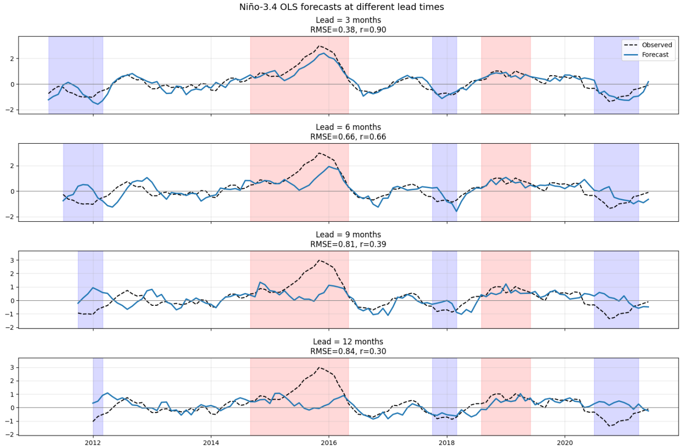
Regression: Fit the Niño 3.4 index curve with different lead times.

Limited number of observations (~ 500 per dataset)
$\rightarrow$ we focus on simple models:

OLS - Linear Regression:

- Good interpretability;
- Works well with the number of observations we have.
- However:
  - Unstable with multicollinearity;
  - Sensitive to noise.

Results - OLS

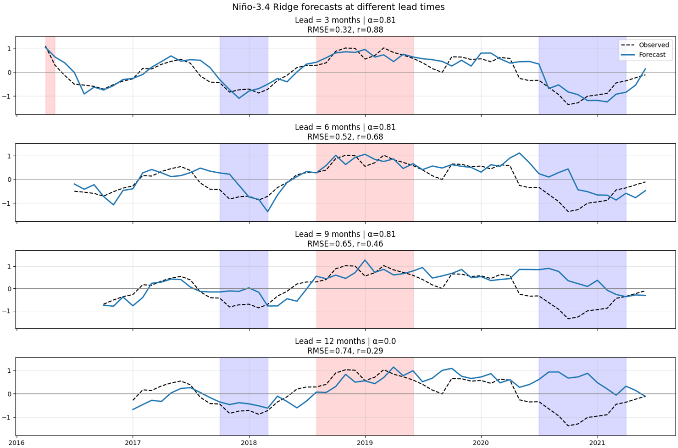

Niño-3.4 OLS forecasts at different lead times
Lead = 3 months
RMSE=0.38, r=0.90
Observed
Forecast
Lead = 6 months
RMSE=0.66, r=0.66
Lead = 9 months
RMSE=0.81, r=0.39
Lead = 12 months
RMSE=0.84, r=0.30
2012
2014
2016
2018
2020

Models: Time series forecasting

Ridge Regression:

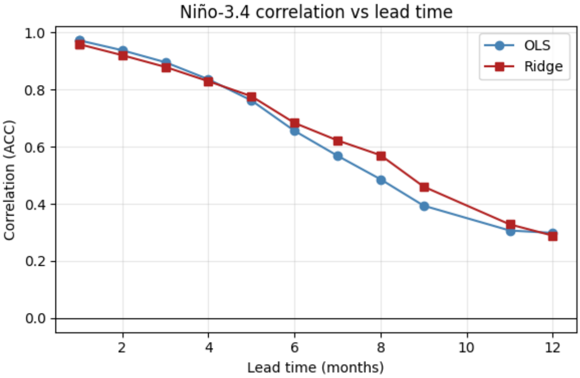
- Handles multicollinearity by shrinking coefficients;
- Reduces variance and overfitting;
- Still linear and relatively interpretable;
- Important:
  - Hyperparameter α must be tuned with grid search;
  - Needs standardization of predictors.

Results - Ridge



Results

Project Question:
How predictable is El Niño? How far ahead can we forecast Niño-3.4 anomalies with simple statistical models?



References

References

[1] Kevin E Trenberth. The definition of el nino. *Bulletin of the American Meteorological Society*, 78(12):2771–2778, 1997.

Thank you !

*Appendix for EOF/PC (1)

Suppose you have a climate variable X:

- Defined on a grid: $(x_i, y_i), \quad i = 1, \ldots, N$
- Observed in time: $\quad t = 1, \ldots, T$

We compute the anomalies:

$$X'(x_i, y_i, t) = X(x_i, y_i, t) - \overline{X}_{\,30\mathrm{y}}(x_i, y_i)$$

where $\overline{X}_{\,30\mathrm{y}}$ denotes the 30-year climatological mean.

*Appendix for EOF/PC (2)

Reshape into a matrix:

$$\mathbf{X} = \begin{bmatrix} X_1'(t_1) & X_1'(t_2) & \cdots & X_1'(t_T) \\ X_2'(t_1) & X_2'(t_2) & \cdots & X_2'(t_T) \\ \vdots & \vdots & \ddots & \vdots \\ X_N'(t_1) & X_N'(t_2) & \cdots & X_N'(t_T) \end{bmatrix}$$

- **Rows:** spatial points
- **Columns:** time

Covariance Matrix: $\mathbf{C} = \frac{1}{T-1}\mathbf{X}\mathbf{X}^\top$

- Equivalent of doing PCA where time samples are observations

*Appendix for EOF/PC (3)

Compute eigenvalues:

$$\mathbf{C}\,\mathbf{e}_k = \lambda_k\,\mathbf{e}_k$$

- $\mathbf{e}_k = \mathrm{EOF}_k$ (spatial pattern)
- λ_k is the variance explained by each mode

Projecting $X(x,y,t)$ into the EOF pattern:

$$\mathrm{PC}_k(t) = \mathbf{e}_k^\top\,\mathbf{X}(:,t)$$

- Each row is a PC time series

*Appendix for Linear Regression

For a given lead time τ, the forecasting is written as:

$$\hat{y}(t+\tau) = \beta_0 + \sum_{i=1}^{N} \beta_i \, X_i(t) + \varepsilon(t)$$

where:

- $\hat{y}(t+\tau)$: predicted Niño-3.4 SST anomaly at lead time τ,
- $X_i(t)$: predictor variables evaluated at time t (lagged Niño indices, SST and MSLP PCs, SOI, and seasonal indicators)
- β_i: regression coefficients estimated from the training data,
- $\varepsilon(t)$ represents unresolved variability and noise.

*Appendix for Ridge Regression

For a given lead time τ, the Ridge regression model is written the same as before, but now the regression coefficients are estimated by minimizing the following cost function:

$$\mathcal{J}(\boldsymbol{\beta}) = \sum_{t} \left[y(t+\tau) - \beta_0 - \sum_{i=1}^{N} \beta_i \, X_i(t) \right]^2 + \alpha \sum_{i=1}^{N} \beta_i^2,$$

- $X_i(t)$: predictor variables at time t (lagged Niño indices, SST and MSLP PCs, SOI, and seasonal indicators)
- β_i: regression coefficients,
- $\varepsilon(t)$: residual variability,
- α: regularization parameter controlling the strength of the penalty (selected using validation period).